ıDEA: an open-source Python framework for exact many-electron quantum mechanics and density functional theory in one dimension

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Abstract

We present ıDEA (interacting Dynamic Electrons Approach), a free and open-source Python framework for the exact solution of the static and time-dependent many-electron Schrödinger equation for finite one-dimensional systems, together with a family of approximate methods (non-interacting, Hartree, Hartree–Fock, local density approximation, hybrid functionals and Kohn–Sham strictly-correlated electrons) that share a common interface. Exact Kohn–Sham, and more generally exact fictitious-system, potentials are obtained by reverse engineering the exact ground-state or time-dependent density. ıDEA is designed for reproducibility, interactivity and simplicity. It has been used to provide exact benchmarks for density functional and many-body perturbation theory, to generate training data for machine learning, and to teach many-body quantum physics. We describe the methods and their implementation, validate them against analytic results, and illustrate their use with three examples.

Keywords: exact diagonalisation, density functional theory, time-dependent DFT, exact Kohn–Sham potential, reverse engineering, one-dimensional model systems

Program summary

Program title: ıDEA (interacting Dynamic Electrons Approach)
CPC Library link to program files: (to be added by Technical Editor)
Developer's repository link: `https://github.com/JackWetherell/iDEA`
Licensing provisions: GNU General Public License v3.0
Programming language: Python ($\geq$ 3.8)
External routines/libraries: NumPy [1], SciPy [2], Matplotlib, tqdm; optionally CuPy [3] for GPU acceleration.
Nature of problem: Exact ground, excited and time-dependent many-electron wavefunctions of finite one-dimensional systems with arbitrary external potentials, interactions and spin configurations are needed as reference data against which approximate methods, such as Kohn–Sham density functional theory (DFT), time-dependent DFT and Hartree–Fock theory, can be assessed. The exact Kohn–Sham potential corresponding to a given interacting density is also required.
Solution method: The many-electron Hamiltonian is represented on a real-space grid using high-order finite differences, and the lowest eigenstates are found by sparse Lanczos diagonalisation followed by explicit antisymmetrisation of the spin-orbital wavefunction. Time propagation uses the action of the matrix exponential. Approximate methods share a self-consistent single-particle framework. Exact Kohn–Sham potentials are found by iteratively inverting the ground-state or time-dependent density.
Additional comments: Installable via `pip install iDEA-latest`. Documentation and a tutorial notebook are provided with the source code.

1. Introduction

Approximate electronic-structure methods are routinely assessed against more accurate methods, but in three dimensions exact many-body solutions are unavailable for all but the smallest systems. Finite one-dimensional (1D) model systems offer a practical alternative: with a softened Coulomb interaction [4] they show many of the qualitative features of real atoms, molecules and materials [5, 6, 7], yet the many-electron Schrödinger equation can be solved exactly for a few electrons. Exact solutions give access to the quantities that approximate theories try to model, most notably the exact Kohn–Sham (KS) potential of density functional theory (DFT) [8, 9] and its time-dependent counterpart [10, 11, 12]. These are unknown in general and must be obtained by inverting the exact density [13, 14, 15, 16, 17].

ıDEA (interacting Dynamic Electrons Approach) grew out of this programme in the group of R. W. Godby at the University of York. Its exact time-dependent solutions and reverse-engineered KS potentials revealed the static and dynamic steps of the exact exchange–correlation (xc) potential [18, 19, 20, 21], complementing related studies by others [22, 23]. The code has since been used to construct local density approximations (LDAs) from finite systems [24, 25], and to assess hybrid functionals [26], the GW approximation [27], the nearsightedness of many-body perturbation theory relative to KS DFT [28], generalized KS theory [29], and exact potentials for the fundamental

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gap [30]. It has also generated exact one-body density matrices for machine learning [31, 32, 33], informed community discussions on excited states and large-scale simulation [34, 35], and been used in teaching.

The current version, iDEA 1.1, is a rewrite of the original research code. It is a pip-installable Python package with a small, uniform interface designed for reproducibility, interactivity (for example in Jupyter notebooks) and simplicity. General-purpose packages either target three-dimensional systems with approximate methods [36, 37], treat lattice or second-quantised models [38, 39], or invert densities supplied by other codes [17]. In contrast, iDEA combines exact real-space many-electron solutions, the standard approximate methods and density inversion, ground state and time dependent, within one consistent framework.

2. Methods and implementation

Systems. A `System` is defined by a uniform grid x_i with spacing δx, an external potential $v_{\text{ext}}(x)$, an interaction $v_{\text{int}}(x, x')$ and a string of electron spins, e.g. `"uud"`. The default interaction is the softened Coulomb interaction $v_{\text{int}}(x, x') = 1/(|x - x'| + 1)$ (Hartree atomic units throughout). The kinetic energy $-\frac{1}{2}\,\mathrm{d}^2/\mathrm{d}x^2$ is represented by central finite differences of 3 to 13 points (default 13), and hard-wall boundary conditions are implied at the grid edges.

Exact solution. For N electrons the spatial Hamiltonian

$$\hat{H} = \sum_{i=1}^{N} \hat{h}_i + \sum_{i<j} v_{\text{int}}(x_i, x_j), \qquad \hat{h} = -\tfrac{1}{2}\tfrac{\mathrm{d}^2}{\mathrm{d}x^2} + v_{\text{ext}}(x), \tag{1}$$

is assembled as a sparse matrix of dimension N_x^N. The one-body part is a Kronecker sum, $\sum_i I\otimes\cdots\otimes h\otimes\cdots\otimes I$, and the interaction is diagonal. The lowest eigenpairs are found with the implicitly restarted Lanczos method (ARPACK [40] via SciPy [2]). Each spatial eigenvector ψ is combined with the spin function χ defined by the spin string, and the antisymmetriser $\sum_P (-1)^P \hat{P}$ is applied explicitly over all $N!$ permutations of the space–spin coordinates. Spatial states with the wrong exchange symmetry vanish and are discarded, so the kth surviving state is the kth fermionic eigenstate. Ground and excited states are thus obtained without imposing symmetry on the spatial problem.

Time propagation in the presence of a perturbation $\delta v(x, t)$ uses $\Psi(t + \delta t) = \exp[-\mathrm{i}\,\delta t\,\hat{H}(t + \delta t)]\,\Psi(t)$, where the action of the matrix exponential is evaluated without forming it, using the algorithm of Al-Mohy and Higham [41]. Optionally, Hamiltonian construction, diagonalisation and propagation run on a GPU through CuPy [3]. On an NVIDIA RTX 4090 this reduces the two-electron ground-state solve on a 450-point grid from about 260 s on an Intel i9-11900K CPU to about 17 s, and 20 propagation steps on a 300-point grid from 3.3 s to 0.5 s (benchmark scripts in the repository).

Approximate methods. Non-interacting electrons, Hartree, restricted and unrestricted Hartree–Fock (HF) [42], the LDA, hybrid functionals and Kohn–Sham strictly-correlated electrons (KS-SCE) [43, 44, 45, 46] share one self-consistent single-particle solver. For each spin channel the Hamiltonian $h_\sigma[n, \rho_\sigma]$ is diagonalised densely, and the density n and spin-resolved density matrices $\rho_\sigma(x, x')$ are mixed linearly until $\int |n - n_{\text{old}}|\,\mathrm{d}x$ falls below a tolerance. Each method supplies only its potential, v_{H}, $v_{\text{H}} + v_{\text{x}}^{\text{HF}}$, $v_{\text{H}} + v_{\text{xc}}^{\text{LDA}}$, $v_{\text{H}} + \alpha v_{\text{x}}^{\text{HF}} + (1-\alpha) v_{\text{xc}}^{\text{LDA}}$, or v_{SCE}, and its total-energy expression. The LDA uses the exchange and correlation energies of the spin-polarised 1D homogeneous electron gas parametrised in Refs. [24, 25]. Excited Slater determinants are obtained by occupying higher orbitals. Time-dependent versions (TDHF, adiabatic LDA, etc.) propagate the occupied orbitals with the same exponential integrator, rebuilding the Hamiltonian from the density and density matrix at each step.

Reverse engineering. For a target density $n_{\text{T}}(x)$ and a chosen method, iDEA finds the external potential of the fictitious system that reproduces n_{T} by iterating

$$v^{(i+1)}(x) = v^{(i)}(x) + \mu\left[n^{(i)}(x)^p - n_{\text{T}}(x)^p\right], \tag{2}$$

until $\int |n^{(i)} - n_{\text{T}}|\,\mathrm{d}x$ is below a tolerance. Here $\mu > 0$ and $0 < p \leq 1$ (default $p = 0.1$, which accelerates convergence where the density is small). With the non-interacting method and an exact n_{T} this yields the exact KS potential v_{KS}, from which $v_{\text{xc}} = v_{\text{KS}} - v_{\text{ext}} - v_{\text{H}}$. In the time-dependent case the potential at each time step is found by solving for the $\delta v(x, t)$ for which a single propagation step reproduces $n_{\text{T}}(x, t)$, using Powell's hybrid method in SciPy.

Observables and interface. The `observables` module computes densities, density matrices, and Hartree and exchange potentials and energies from many-body states, single-particle states or their time evolution. Every method module exposes the same functions (`solve`, `propagate`, `total_energy`, `hamiltonian`), so methods can be swapped or looped over:

```
import numpy as np, iDEA
x = np.linspace(-12, 12, 241)
s = iDEA.system.System(x, -2/(abs(x)+1),
        iDEA.interactions.softened_interaction(x),
        electrons="uu")
gs = iDEA.methods.interacting.solve(s, k=0)
n = iDEA.observables.density(s, state=gs)
for m in iDEA.iterate_sb_methods:   # NI, H, HF, LDA, hybrid
    st = m.solve(s)
    print(m.name, m.total_energy(s, st))
ks = iDEA.reverse_engineering.reverse(
        s, n, iDEA.methods.non_interacting)
v_xc = ks.v_ext - s.v_ext \
       - iDEA.observables.hartree_potential(s, n)
```

3. Validation

The exact solver is tested against a semi-analytic reference (Table 1). For two electrons in a harmonic potential $v_{\text{ext}} = \frac{1}{2}\omega^2 x^2$ with the softened interaction, the Hamiltonian separates into an analytic centre-of-mass part and a relative-coordinate problem, which is solved on a fine grid. The iDEA energies, densities and wavefunctions agree with this reference for both the spin-singlet (`ud`) and spin-triplet (`uu`) states. The harmonic-potential theorem [47] provides a second test. The lowest dipole

Table 1: Validation of the exact solver and indicative cost. (a) Two electrons in a harmonic well ($\omega = 0.3275$, $N_x = 231$, $L = 20$) compared with the separated centre-of-mass/relative-coordinate reference on the same grid: absolute energy error, maximum density error and wavefunction infidelity $1-|\langle\Psi_{\mathrm{ref}}|\Psi\rangle|$. (b) Lowest dipole excitation energy versus the harmonic-potential theorem, and the time-dependent density error over 500 steps. (c) Grid convergence of the ground-state energy of the atom in Fig. 1. (d) Wall time of `interacting.solve` on 4 CPU cores of the test machine.

(a) Two-electron ground state	singlet (ud)	triplet (uu)
E (Ha)	0.779893	0.936028
$\lvert\Delta E\rvert$	3.7×10^{-12}	1.6×10^{-12}
$\max\lvert\Delta n\rvert$	1.1×10^{-11}	4.0×10^{-12}
infidelity	5.8×10^{-15}	3.7×10^{-14}
(b) Excitations and dynamics		
$E_1 - E_0$ vs $\omega = 0.25$	0.25000004	error 4.0×10^{-8}
TD density, $\max\lvert\Delta n(x,t)\rvert$	2.0×10^{-13}	
(c) Grid convergence, atom (uu)		
$\delta x = 0.200$	$E = -1.34519$	
$\delta x = 0.133$	$E = -1.34131$	
$\delta x = 0.100$	$E = -1.33995$	
(d) Ground-state solve time		
$N = 2$, $N_x = 60$	0.2 s	
$N = 2$, $N_x = 100$	1.4 s	
$N = 2$, $N_x = 140$	3.4 s	
$N = 2$, $N_x = 180$	9.7 s	
$N = 2$, $N_x = 220$	21.5 s	
$N = 3$, $N_x = 20$	1.1 s	
$N = 3$, $N_x = 30$	6.9 s	
$N = 3$, $N_x = 40$	19.7 s	

excitation of interacting electrons in a harmonic well has energy exactly ω, which IDEA reproduces. For time propagation, a single electron displaced in a harmonic well and released undergoes coherent oscillation with an analytic density, which the propagator reproduces at every time step. Because the reference uses the same finite-difference stencil, these tests isolate the many-body construction, eigensolver and antisymmetrisation from discretisation error, which is controlled separately by the grid spacing: for the atom of Fig. 1, whose potential has a cusp at the origin, the ground-state energy changes by about 10^{-3} Ha between $\delta x = 0.133$ and 0.100 (Table 1c). These tests run in continuous integration, and further tests check the self-consistent methods and a reverse-engineering round trip.

4. Examples

All results below are produced by `paper/scripts/make_figures.py` in the repository, which regenerates every figure and number in this paper.

Ground state: comparing approximations. Figure 1 shows two spin-polarised electrons bound by a softened nucleus of charge 2. The exact ground-state energy is $E = -1.3413$ Ha. HF (-1.3388 Ha), the LDA (-1.3348 Ha) and the hybrid (-1.3379 Ha) are compared with Hartree theory (-0.8852 Ha) and non-interacting electrons (-1.6388 Ha). The integrated density errors $\int |n - n_{\mathrm{exact}}|\,\mathrm{d}x$ are 0.005 (HF), 0.050 (LDA),

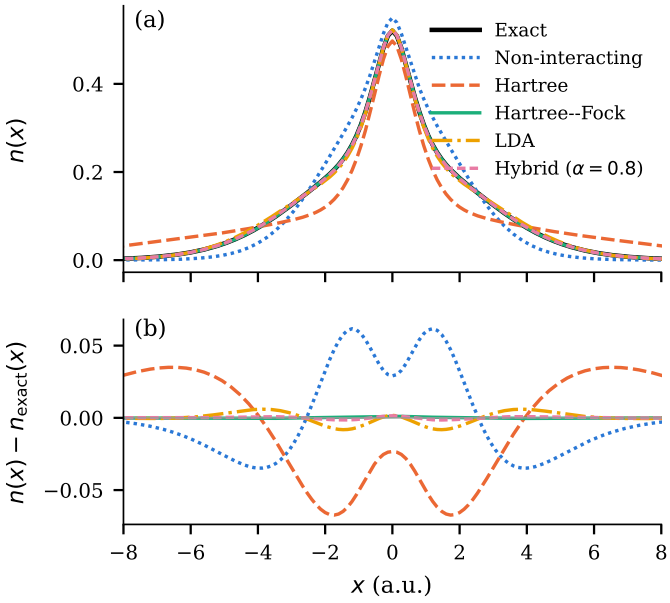


Figure 1: (a) Ground-state density of two spin-polarised electrons in the softened atom $v_{\mathrm{ext}} = -2/(|x| + 1)$ ($N_x = 181$, $\delta x = 0.133$), exact and from each approximate method. (b) Error relative to the exact density.

0.009 (hybrid) and 0.654 (Hartree). The self-interaction contained in the Hartree potential makes its density far too diffuse. The LDA removes much of this error but still slightly over-delocalises the electrons [48], while HF, which is free of self-interaction for this system, gives the most accurate density. Each method is obtained with the same call, so a comparison like this is a few lines of code.

Exact Kohn–Sham potentials. Reverse engineering the exact density of the spin-polarised atom (Fig. 2a) gives the exact KS potential, and hence v_{xc}. The density of the resulting KS system reproduces the exact density to $\int |n_{\mathrm{KS}} - n|\,\mathrm{d}x = 1.0\times10^{-8}$. Near the nucleus the exact v_{xc} and the LDA evaluated on the same density are close. In the low-density tail, however, the LDA decays too quickly: at $x = -8$, where $n \approx 3\times10^{-3}$, the exact v_{xc} is -0.060 Ha but the LDA gives -0.029 Ha. The exact potential must decay slowly there to cancel the self-interaction contained in the Hartree potential. For the spin-singlet atom (Fig. 2b, exact energy -1.7162 Ha) the exact exchange potential of two electrons in one spatial orbital is $-v_{\mathrm{H}}/2$, so reverse engineering isolates the exact correlation potential v_{c} directly. The same procedure, applied with other methods in place of the non-interacting solver, gives exact Hartree- or HF-like fictitious systems.

Dynamics. Figure 3 shows the free dipole oscillation of the same atom after a short field pulse. The exact many-body propagation of 1000 steps takes about 80 s on the test machine. The approximate methods are propagated through the same interface. The adiabatic LDA oscillates too fast and time-dependent Hartree theory too slowly, while TDHF and the hybrid follow the exact dipole closely over the first period. The exact time-dependent density can in turn be reverse engineered to give the

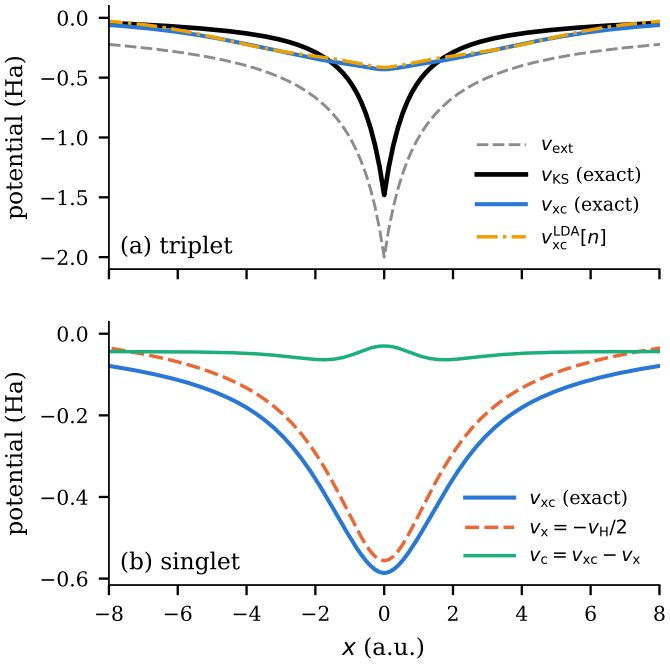


Figure 2: Exact KS and xc potentials obtained by reverse engineering exact densities of the softened atom of Fig. 1. (a) Spin-polarised (uu) electrons, compared with the LDA xc potential of the same density. (b) Spin-singlet (ud) electrons, for which the exact exchange potential is $-v_H/2$, giving the exact correlation potential v_c. Potentials are shifted to vanish at the left edge of the box.

exact time-dependent KS potential [19, 20].

5. Conclusions

iDEA provides exact many-electron solutions of 1D model systems, the standard approximate methods and exact reverse engineering of KS and other fictitious-system potentials, ground state and time dependent, behind a single small Python interface. Because the wavefunction is stored on an N_x^N grid, the exact solver is limited in practice to two or three electrons on grids of a few hundred points (Table 1). This is enough for the reference calculations that motivate the code and allows them to run interactively. The simple density-update scheme of Eq. (2) converges reliably for single systems, but it can stall for weakly coupled subsystems, where the KS orbitals are nearly degenerate. More robust inversion algorithms [15, 16, 17] are a natural extension. Future directions include symmetry-adapted and tensor-network representations to reach more electrons, and further approximate functionals. iDEA is developed openly, and contributions are welcome.

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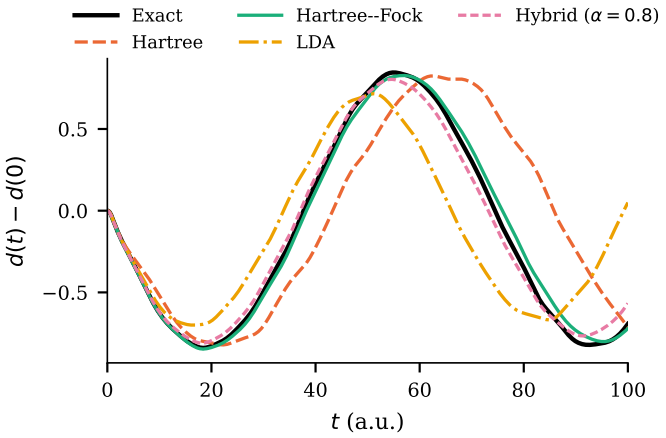


Figure 3: Dipole moment $d(t) = \int x\, n(x,t)\, dx$ of the atom of Fig. 1 after a uniform-field pulse of strength 0.05 a.u. and duration 1 a.u., from exact propagation and from the time-dependent Hartree, HF, adiabatic LDA and hybrid methods ($N_x = 121$, 1000 time steps).

References

[1] C. R. Harris, K. J. Millman, S. J. van der Walt, R. Gommers, P. Virtanen, D. Cournapeau, E. Wieser, J. Taylor, S. Berg, N. J. Smith, R. Kern, M. Picus, S. Hoyer, M. H. van Kerkwijk, M. Brett, A. Haldane, J. Fernández del Río, M. Wiebe, P. Peterson, P. Gérard-Marchant, K. Sheppard, T. Reddy, W. Weckesser, H. Abbasi, C. Gohlke, T. E. Oliphant, Array programming with NumPy, Nature 585 (7825) (2020) 357–362. doi:10.1038/s41586-020-2649-2.

[2] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, İ. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, SciPy 1.0 Contributors, SciPy 1.0: Fundamental algorithms for scientific computing in Python, Nat. Methods 17 (3) (2020) 261–272. doi:10.1038/s41592-019-0686-2.

[3] R. Okuta, Y. Unno, D. Nishino, S. Hido, C. Loomis, CuPy: A NumPy-compatible library for NVIDIA GPU calculations, in: Proceedings of Workshop on Machine Learning Systems (LearningSys) in The Thirty-first Annual Conference on Neural Information Processing Systems (NIPS), 2017.
URL http://learningsys.org/nips17/assets/papers/paper_16.pdf

[4] J. Javanainen, J. H. Eberly, Q. Su, Numerical simulations of multiphoton ionization and above-threshold electron spectra, Phys. Rev. A 38 (7) (1988) 3430–3446. doi:10.1103/PhysRevA.38.3430.

[5] L. O. Wagner, E. M. Stoudenmire, K. Burke, S. R. White, Reference electronic structure calculations in one dimension, Phys. Chem. Chem. Phys. 14 (24) (2012) 8581–8590. doi:10.1039/c2cp24118h.

[6] T. E. Baker, E. M. Stoudenmire, L. O. Wagner, K. Burke, S. R. White, One-dimensional mimicking of electronic structure: The case for exponentials, Phys. Rev. B 91 (2015) 235141. doi:10.1103/PhysRevB.91.235141.

[7] N. Helbig, J. I. Fuks, M. Casula, M. J. Verstraete, M. A. L. Marques, I. V. Tokatly, A. Rubio, Density functional theory beyond the linear regime: Validating an adiabatic local density approximation, Phys. Rev. A 83 (2011) 032503. doi:10.1103/PhysRevA.83.032503.

[8] P. Hohenberg, W. Kohn, Inhomogeneous electron gas, Phys. Rev. 136 (1964) B864–B871. doi:10.1103/PhysRev.136.B864.

[9] W. Kohn, L. J. Sham, Self-consistent equations including exchange and correlation effects, Phys. Rev. 140 (1965) A1133–A1138. doi:10.1103/PhysRev.140.A1133.

[10] E. Runge, E. K. U. Gross, Density-functional theory for time-dependent systems, Phys. Rev. Lett. 52 (1984) 997–1000. doi:10.1103/PhysRevLett.52.997.

[11] N. T. Maitra, Perspective: Fundamental aspects of time-dependent density functional theory, J. Chem. Phys. 144 (2016) 220901. doi:10.1063/1.4953039.

[12] C. A. Ullrich, Time-Dependent Density-Functional Theory: Concepts and Applications, Oxford University Press, Oxford, 2012.

[13] R. van Leeuwen, E. J. Baerends, Exchange-correlation potential with correct asymptotic behavior, Phys. Rev. A 49 (1994) 2421–2431. doi:10.1103/PhysRevA.49.2421.

[14] Q. Wu, W. Yang, A direct optimization method for calculating density functionals and exchange–correlation potentials from electron densities, J. Chem. Phys. 118 (6) (2003) 2498–2509. doi:10.1063/1.1535422.

[15] D. S. Jensen, A. Wasserman, Numerical methods for the inverse problem of density functional theory, Int. J. Quantum Chem. 118 (1) (2018) e25425. doi:10.1002/qua.25425.

[16] B. Kanungo, P. M. Zimmerman, V. Gavini, Exact exchange-correlation potentials from ground-state electron densities, Nat. Commun. 10 (2019) 4497. doi:10.1038/s41467-019-12467-0.

[17] Y. Shi, V. H. Chávez, A. Wasserman, n2v: A density-to-potential inversion suite. A sandbox for creating, testing, and benchmarking density functional theory inversion methods, WIREs Comput. Mol. Sci. 12 (6) (2022) e1617. doi:10.1002/wcms.1617.

[18] J. D. Ramsden, R. W. Godby, Exact density-functional potentials for time-dependent quasiparticles, Phys. Rev. Lett. 109 (3) (2012) 036402. doi:10.1103/PhysRevLett.109.036402.

[19] M. J. P. Hodgson, J. D. Ramsden, J. B. J. Chapman, P. Lillystone, R. W. Godby, Exact time-dependent density-functional potentials for strongly correlated tunneling electrons, Phys. Rev. B 88 (24) (2013) 241102(R). doi:10.1103/PhysRevB.88.241102.

[20] M. J. P. Hodgson, J. D. Ramsden, R. W. Godby, Origin of static and dynamic steps in exact Kohn-Sham potentials, Phys. Rev. B 93 (15) (2016) 155146. doi:10.1103/PhysRevB.93.155146.

[21] M. J. P. Hodgson, E. Kraisler, A. Schild, E. K. U. Gross, How interatomic steps in the exact Kohn-Sham potential relate to derivative discontinuities of the energy, J. Phys. Chem. Lett. 8 (24) (2017) 5974–5980. doi:10.1021/acs.jpclett.7b02615.

[22] P. Elliott, J. I. Fuks, A. Rubio, N. T. Maitra, Universal dynamical steps in the exact time-dependent exchange-correlation potential, Phys. Rev. Lett. 109 (26) (2012) 266404. doi:10.1103/PhysRevLett.109.266404.

[23] M. Ruggenthaler, M. Penz, R. van Leeuwen, Existence, uniqueness, and construction of the density-potential mapping in time-dependent density-functional theory, J. Phys.: Condens. Matter 27 (2015) 203202. doi:10.1088/0953-8984/27/20/203202.

[24] M. T. Entwistle, M. J. P. Hodgson, J. Wetherell, B. Longstaff, J. D. Ramsden, R. W. Godby, Local density approximations from finite systems, Phys. Rev. B 94 (20) (2016) 205134. doi:10.1103/PhysRevB.94.205134.

[25] M. T. Entwistle, M. Casula, R. W. Godby, Comparison of local density functionals based on electron gas and finite systems, Phys. Rev. B 97 (23) (2018) 235143. doi:10.1103/PhysRevB.97.235143.

[26] A. R. Elmaslmane, J. Wetherell, M. J. P. Hodgson, K. P. McKenna, R. W. Godby, Accuracy of electron densities obtained via Koopmans-compliant hybrid functionals, Phys. Rev. Materials 2 (4) (2018) 040801(R). doi:10.1103/PhysRevMaterials.2.040801.

[27] J. Wetherell, M. J. P. Hodgson, R. W. Godby, GW self-screening error and its correction using a local density functional, Phys. Rev. B 97 (12) (2018) 121102(R). doi:10.1103/PhysRevB.97.121102.

[28] J. Wetherell, M. J. P. Hodgson, L. Talirz, R. W. Godby, Advantageous nearsightedness of many-body perturbation theory contrasted with Kohn-Sham density functional theory, Phys. Rev. B 99 (4) (2019) 045129. doi:10.1103/PhysRevB.99.045129.

[29] M. J. P. Hodgson, J. Wetherell, Accurate real-time evolution of electron densities and ground-state properties from generalized Kohn-Sham theory, Phys. Rev. A 101 (3) (2020) 032502. doi:10.1103/PhysRevA.101.032502.

[30] M. J. P. Hodgson, J. Wetherell, E. Fromager, Exact exchange-correlation potentials for calculating the fundamental gap with a fixed number of electrons, Phys. Rev. A 103 (1) (2021) 012806. doi:10.1103/PhysRevA.103.012806.

[31] J. Wetherell, A. Costamagna, M. Gatti, L. Reining, Insights into one-body density matrices using deep learning, Faraday Discuss. 224 (2020) 265–291. doi:10.1039/D0FD00061B.

[32] J. C. Snyder, M. Rupp, K. Hansen, K.-R. Müller, K. Burke, Finding density functionals with machine learning, Phys. Rev. Lett. 108 (25) (2012) 253002. doi:10.1103/PhysRevLett.108.253002.

[33] L. Li, J. C. Snyder, I. M. Pelaschier, J. Huang, U.-N. Niranjan, P. Duncan, M. Rupp, K.-R. Müller, K. Burke, Understanding machine-learned density functionals, Int. J. Quantum Chem. 116 (2016) 819–833. doi:10.1002/qua.25040.

[34] J. G. Brandenburg, et al., New approaches to study excited states in density functional theory: general discussion, Faraday Discuss. 224 (2020) 483–508. doi:10.1039/D0FD90026E.

[35] J. G. Brandenburg, et al., Challenges for large scale simulation: general discussion, Faraday Discuss. 224 (2020) 309–332. doi:10.1039/D0FD90024A.

[36] N. Tancogne-Dejean, M. J. T. Oliveira, et al., Octopus, a computational framework for exploring light-driven phenomena and quantum dynamics in extended and finite systems, J. Chem. Phys. 152 (2020) 124119. doi:10.1063/1.5142502.

[37] Q. Sun, et al., Recent developments in the PySCF program package, J. Chem. Phys. 153 (2020) 024109. doi:10.1063/5.0006074.

[38] P. Weinberg, M. Bukov, QuSpin: a Python package for dynamics and exact diagonalisation of quantum many body systems part I: spin chains, SciPost Phys. 2 (2017) 003. doi:10.21468/SciPostPhys.2.1.003.

[39] J. R. McClean, et al., OpenFermion: the electronic structure package for quantum computers, Quantum Sci. Technol. 5 (2020) 034014. doi:10.1088/2058-9565/ab8ebc.

[40] R. B. Lehoucq, D. C. Sorensen, C. Yang, ARPACK Users' Guide: Solution of Large-Scale Eigenvalue Problems with Implicitly Restarted Arnoldi Methods, SIAM, Philadelphia, 1998. doi:10.1137/1.9780898719628.

[41] A. H. Al-Mohy, N. J. Higham, Computing the action of the matrix exponential, with an application to exponential integrators, SIAM J. Sci. Comput. 33 (2) (2011) 488–511. doi:10.1137/100788860.

[42] V. Fock, Näherungsmethode zur Lösung des quantenmechanischen Mehrkörperproblems, Z. Phys. 61 (1930) 126–148. doi:10.1007/BF01340294.

[43] F. Malet, P. Gori-Giorgi, Strong correlation in Kohn-Sham density functional theory, Phys. Rev. Lett. 109 (24) (2012) 246402. doi:10.1103/PhysRevLett.109.246402.

[44] F. Malet, A. Mirtschink, J. C. Cremon, S. M. Reimann, P. Gori-Giorgi, Kohn-Sham density functional theory for quantum wires in arbitrary correlation regimes, Phys. Rev. B 87 (2013) 115146. doi:10.1103/PhysRevB.87.115146.

[45] A. Mirtschink, M. Seidl, P. Gori-Giorgi, Derivative discontinuity in the strong-interaction limit of density-functional theory, Phys. Rev. Lett. 111 (2013) 126402. doi:10.1103/PhysRevLett.111.126402.

[46] A. Marie, D. P. Kooi, J. Grossi, M. Seidl, Z. H. Musslimani, K. J. H. Giesbertz, P. Gori-Giorgi, Real-space Mott-Anderson electron localization with long-range interactions, Phys. Rev. Research 4 (4) (2022) 043192. doi:10.1103/PhysRevResearch.4.043192.

[47] J. F. Dobson, Harmonic-potential theorem: Implications for approximate many-body theories, Phys. Rev. Lett. 73 (16) (1994) 2244–2247. doi:10.1103/PhysRevLett.73.2244.

[48] P. Mori-Sánchez, A. J. Cohen, W. Yang, Localization and delocalization errors in density functional theory and implications for band-gap prediction, Phys. Rev. Lett. 100 (2008) 146401. doi:10.1103/PhysRevLett.100.146401.